Information limits and temporal design for ghost imaging through dynamic scattering media

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Abstract

Ghost imaging through a dynamic scattering medium replaces a fixed forward operator with a time-varying gain, and correcting for that gain is the central obstacle to quantitative reconstruction. We derive an exact missing-information identity showing that a hidden multiplicative gain path removes precisely the conditional covariance of the complete object score given the bucket record. In the Poisson bucket channel this loss specializes to the posterior covariance of a hidden gain-weighted illumination vector, which yields an estimator-independent Bayes-risk ceiling on any correction — neural or otherwise — under the declared gain model and prior. A Schur-complement analysis then characterizes the information-optimal temporal schedules: chopping, interleaving, and randomization help exactly when they make the object score orthogonal, in the noise metric, to the admitted gain-drift modes. All statements are conditioned on a stated scalar-gain hypothesis and its physical validity boundary; a resource-conditioned temporal operating surface is left as a conditional program rather than a universal claim.

1 Introduction

Single-pixel and computational ghost imaging (GI) reconstruct a scene from the responses of one bucket detector to a sequence of structured illumination patterns [1, 2, 3]. When the light passes through a dynamic scattering medium, the effective forward operator acquires a time-varying gain, and a substantial experimental line corrects for that gain [4, 5, 6, 7, 8]. That literature establishes methods and experiments; it does not establish an exact observed-versus-complete Fisher decomposition, an estimator-independent Bayes-risk ceiling, or an information-optimal temporal schedule under correlated gain. Those are the residual questions addressed here.

This manuscript is the information-and-design companion to Part I [9], which asks when the object and gain class can be separated at all. Part I settles identifiability; the present paper asks, once separation is possible, how much object information survives hidden gain, what estimator-independent risk floor follows, and how patterns should be ordered in time. The two are companion questions, not sections of one paper, and are not merged. The argument follows a single hierarchy in which each arrow requires its own assumptions and none may be skipped:

scalar-gain hypothesis $\rightarrow$ identifiability $\rightarrow$ observed information $\rightarrow$ Bayes risk bound $\rightarrow$ temporal design.

2 Physical hypothesis MG and channel model

The entire program rests on a physical hypothesis: that the dynamic medium acts, on the illuminated object subspace, as a single scalar multiplier per frame plus a controlled residual. This section states the hypothesis, its sufficient physical conditions, and the regimes in which it breaks.

Hypothesis 1 (MG — scalar multiplicative gain)**.** *Write the dynamic medium's intensity-transfer functional at time t as $\mathcal{K}_t$. The scalar multiplicative model is a valid approximation on the experiment's illuminated object subspace $\mathcal{S}$ only if there exist a fixed functional $\mathcal{K}_0$, a positive scalar $a(t)$, and a small residual E_t such that*

$$\mathcal{K}_t|_{\mathcal{S}} = a(t)\,\mathcal{K}_0|_{\mathcal{S}} + E_t, \qquad \|E_t z\| \le \varepsilon\, a(t)\,\|\mathcal{K}_0 z\| \quad \forall z \in \mathcal{S}. \tag{MG}$$

After frame integration, a_n is either the quasi-static gain or the properly weighted frame average.

Plausible sufficient physical conditions for Hypothesis 1 are:

1. the bucket collects enough output speckles/modes that temporal medium variation mainly changes total throughput rather than the normalized spatial transfer;
2. the object and medium are static enough within a pattern exposure for a scalar frame average to be meaningful;
3. the pattern does not perturb the medium;
4. incoherent/intensity-linear propagation is an adequate description, or coherent cross terms average out at the bucket;
5. additive scattered background is separately modeled;
6. the same collection aperture, polarization, and wavelength distribution apply across patterns.

The scalar model fails when:

- the dynamic medium changes the spatial transmission matrix across object coordinates;
- only a few speckles are collected, producing pattern-dependent non-Gaussian fluctuations;
- coherent bucket integration retains interference cross terms that depend on the displayed pattern;
- the medium evolves substantially within a frame and cannot be represented by one scalar average;
- object motion and medium motion couple to the pattern order;
- polarization, wavelength, or path-length composition changes with time;
- multiple scattering creates an additive or convolutive component rather than common multiplicative throughput.

Remark 1 (Scope of the scalar model)*.* In the broken regimes the correct latent object is a vector/matrix-valued transfer process. The general missing-information identity of Section 3 survives, but the low-dimensional correction theory and the Poisson hidden-exposure corollary of Section 4 do not. A second risk is gauge singularity: in a non-identifiable square design both complete and observed information can be singular. The identity remains true but cannot manufacture identifiability, so Part I's thresholds [9] must be assumed before any inverse information or Bayes risk is interpreted.

3 The exact gain-path missing-information theorem

Let X denote the object parameter, L the latent gain path, and Y the observed bucket record. Assume: (1) $p(l)$ does not depend on x; (2) $p(y, l \mid x) = p(l)\, p(y \mid x, l)$ is dominated and differentiable in x; (3) differentiation may pass under the integral; (4) the complete score is square-integrable. Define the complete object score

$$U_x(Y, L) = \nabla_x \log p(Y, L \mid x) = \nabla_x \log p(Y \mid x, L).$$

Theorem 1 (O1 — scattering missing-information identity)**.** *The observed score is*

$$\nabla_x \log p(Y \mid x) = \mathbb{E}\{U_x(Y, L) \mid Y, x\}. \tag{O1.1}$$

The complete and observed Fisher matrices satisfy

$$I_Y(x) = I_{Y,L}(x) - \mathbb{E}_Y[\mathrm{Cov}\{U_x(Y, L) \mid Y, x\}]\,. \tag{O1.2}$$

The missing term is positive semidefinite. It vanishes if and only if the complete object score is measurable from the observed record.

Proof. Differentiate the marginal likelihood:

$$\nabla_x p(y \mid x) = \int \nabla_x p(y, l \mid x)\, dl = \int p(y, l \mid x)\, U_x(y, l)\, dl.$$

Division by $p(y \mid x)$ gives (O1.1). Scores have mean zero, so the law of total covariance gives

$$\mathrm{Cov}(U_x) = \mathrm{Cov}\{\mathbb{E}(U_x \mid Y)\} + \mathbb{E}\{\mathrm{Cov}(U_x \mid Y)\}.$$

The first covariance is the observed Fisher matrix and the left side is the complete Fisher matrix, proving (O1.2). □

This proof is classical Louis theory [10]. The following programmable-GI specialization is the important part.

4 Poisson hidden-exposure corollary and Gaussian counterpart

4.1 Exact Poisson bucket corollary

Let the rows of $M \in \mathbb{R}_+^{N\times K}$ be $m_n^\top$, let

$$s = Mx, \qquad s_n > 0, \qquad a_n = e^{l_n} > 0,$$

and conditionally on the entire correlated gain vector a, let $Y_n \mid a, x \sim \mathrm{Poisson}(a_n s_n)$ independently across frames. The prior on a may have arbitrary temporal correlation and need not be Gaussian. The complete score is

$$U_x = M^\top D_s^{-1} Y - M^\top a. \tag{O1.3}$$

Conditionally on a, $I_{Y|a}(x) = M^\top \operatorname{diag}(a_n/s_n)\, M$.

Corollary 1 (O1-P — hidden exposure-vector identity)**.** *Under the conditionally Poisson multiplicative-gain model,*

$$I_Y(x) = M^\top \operatorname{diag}\left(\frac{\mathbb{E}a_n}{s_n}\right) M - M^\top \mathbb{E}_Y\{\mathrm{Cov}(a \mid Y, x)\}M. \tag{O1.4}$$

Proof. Average the conditional Fisher matrix for the complete-information term. Given Y, the first term of (O1.3) is fixed, hence $\mathrm{Cov}(U_x \mid Y, x) = M^\top \mathrm{Cov}(a \mid Y, x)\, M$. Insert this into (O1.2). □

What plays the role of live time. For the dead-time channel of Part I's dead-time line, hidden active time is the latent exposure to the rate parameter. Here the corresponding object is

$$G_M(a) = M^\top a = \sum_{n=1}^{N} a_n m_n, \tag{O1.5}$$

the *gain-weighted illumination/exposure vector*. The information loss is exactly the posterior covariance of this hidden exposure vector,

$$I_{\text{missing}} = \mathbb{E}_Y \operatorname{Cov}\{G_M(a) \mid Y, x\}.$$

This is the cleanest bridge from Part I's live-time identity to dynamic-scattering GI. Three consequences are immediate:

1. temporal correlation enters through the full posterior covariance of a, not through one scalar variance;
2. pattern ordering matters because it rotates gain uncertainty through $M^\top(\cdot)M$;
3. a second monitor detector or other gain side information can only reduce the missing term by conditioning more strongly.

4.2 Gaussian readout corollary

For analog bucket data $Y \mid a, x \sim \mathcal{N}(D_a M x, R)$ with $R \succ 0$, the complete score and information are

$$U_x = M^\top D_a R^{-1}(Y - D_a M x), \qquad I_{Y,a}(x) = \mathbb{E}_a[\, M^\top D_a R^{-1} D_a M \,].$$

Identity (O1.2) remains exact, but the missing term is

$$\mathbb{E}_Y \operatorname{Cov}\Big[M^\top D_a R^{-1}(Y - D_a M x) \mid Y, x\Big]\,,$$

not merely $M^\top \operatorname{Cov}(a \mid Y)\, M$. The unusually clean hidden-exposure formula (O1.4) is therefore a property of the conditionally Poisson multiplicative channel, not a universal gain law.

5 Universal correction ceiling via van Trees

5.1 Estimator-independent Bayesian bound

Let the object have differentiable prior $\pi(x)$, with prior information $I_\pi = \mathbb{E}_\pi[\nabla \log \pi(X)\, \nabla \log \pi(X)^\top]$. Under the usual van Trees boundary conditions [11, 12], every estimator $\widehat{x}(Y)$ satisfies

$$\mathbb{E}[(\widehat{x} - X)(\widehat{x} - X)^\top] \succeq [I_\pi + \mathbb{E}_X I_Y(X)]^{-1}\,. \tag{O3.1}$$

Combining with Theorem 1 gives

$$\mathbb{E}[(\widehat{x} - X)(\widehat{x} - X)^\top] \succeq [I_\pi + \mathbb{E} I_{Y,L}(X) - \mathbb{E}\operatorname{Cov}(U_X \mid Y, X)]^{-1}\,. \tag{O3.2}$$

For a task matrix $W \succeq 0$,

$$\mathbb{E}\|W^{1/2}(\widehat{x} - X)\|_2^2 \geq \operatorname{tr}\big\{W\, [I_\pi + \mathbb{E} I_Y(X)]^{-1}\big\}\,. \tag{O3.3}$$

This covers a U-Net, an untrained network, a joint MAP estimator, or any other estimator that receives the same bucket record and side information.

Remark 2 (What may be claimed). Under the declared gain prior and observation model, no estimator using the same measurements can achieve Bayes MSE below (O3.3).

Remark 3 (What may not be claimed). The following are outside the scope of (O3.3):

- the bound does not show that a particular network is close to optimal;
- it does not directly bound PSNR, SSIM, visibility, or CNR without an explicit transformation/loss argument;
- if a network receives additional calibration data, reference-detector data, or a mismatched training distribution, those must be included in the experiment and prior;
- a lower bound alone does not prove that "more training is pointless"; that conclusion requires the achieved risk to be close to the bound.

5.2 Connection to Part I's static-correction ceiling

Use the small-log-gain linearization

$$a \simeq \mathbf{1} + \ell, \qquad Y = s + D_s\ell + \epsilon, \qquad s = Mx,$$

with $\ell \sim \mathcal{N}(0, C_\ell)$ and $\epsilon \sim \mathcal{N}(0, R)$. Suppose a static correction removes a declared drift subspace with projector P. The residual gain is $\ell_\perp = (I-P)\ell$ and $C_\perp = (I-P)C_\ell(I-P)^\top$. The marginalized model is

$$Y \mid x \sim \mathcal{N}(s, \Sigma_x), \qquad \Sigma_x = R + D_s C_\perp D_s. \tag{O3.4}$$

Its exact Gaussian Fisher matrix is

$$[I_Y]_{jk} = (\partial_j s)^\top \Sigma_x^{-1}(\partial_k s) + \tfrac{1}{2}\operatorname{tr}\left(\Sigma_x^{-1}\partial_j\Sigma_x\, \Sigma_x^{-1}\partial_k\Sigma_x\right). \tag{O3.5}$$

If $C_\perp = v\bar{C}$ and $v \to 0$, the covariance-derivative term is $O(v^2)$, while

$$I_Y = M^\top R^{-1}M - v\, M^\top R^{-1} D_s \bar{C} D_s R^{-1} M + O(v^2). \tag{O3.6}$$

Remark 4 (The vB_L term — derivation TODO). A scalarized risk or relative-MSE expansion of (O3.6) yields a term of the form vB_L, so Part I's static-correction ceiling can be interpreted as the first-order scalarization of the exact missing-information/Bayes-risk architecture — but only under the small-gain Gaussian model, the declared correction projector, and the chosen task scalarization. The ruling sketches this connection but does not carry out the scalarization. [TODO: derive the vB_L term explicitly: fix the task scalarization W, expand $\operatorname{tr}\{W\,[I_\pi + \mathbb{E}I_Y]^{-1}\}$ from (O3.6) to first order in v, and identify B_L with Part I's static-correction coefficient. This is a theorem path, not an automatic identity between manuscripts.]

6 Nuisance-orthogonal temporal design

6.1 Linearized drift model

Let the gain path be represented locally by $\ell = H\beta + r$, where the columns of $H \in \mathbb{R}^{N\times p}$ are declared low-frequency or OU/KL drift modes, β has prior precision Λ_β, and r is residual gain noise. At $\beta = 0$, use the local Gaussian model

$$y = M_\pi x + D_{s_\pi} H\beta + \epsilon, \qquad s_\pi = M_\pi x, \qquad \epsilon \sim \mathcal{N}(0, R), \tag{O4.1}$$

where π denotes the temporal order/schedule of the programmable patterns. The joint information matrix for (x, β) is

$$\mathcal{I}(\pi) = \begin{bmatrix} I_{xx} & I_{x\beta} \\ I_{\beta x} & I_{\beta\beta} \end{bmatrix},$$

with

$$I_{xx} = M_\pi^\top R^{-1} M_\pi, \quad I_{x\beta} = M_\pi^\top R^{-1} D_{s_\pi} H, \quad I_{\beta\beta} = H^\top D_{s_\pi} R^{-1} D_{s_\pi} H + \Lambda_\beta.$$

The efficient/Bayesian information for the object after treating drift as nuisance is the Schur complement

$$I_{x|\text{gain}}(\pi) = I_{xx}(\pi) - I_{x\beta}(\pi)\, I_{\beta\beta}(\pi)^{-1}\, I_{\beta x}(\pi). \tag{O4.2}$$

6.2 The nuisance-orthogonal schedule

Theorem 2 (O4-A — nuisance-orthogonal schedule)**.** *Assume $I_{\beta\beta} \succ 0$. For schedules having the same oracle object information I_{xx},*

$$I_{x|\text{gain}} \preceq I_{xx},$$

with equality if and only if

$$M_\pi^\top R^{-1} D_{M_\pi x} H = 0. \tag{O4.3}$$

Proof. The subtracted term in (O4.2) is of the form $AA^\top$ with $A = I_{x\beta} I_{\beta\beta}^{-1/2}$, and is therefore positive semidefinite. It vanishes exactly when $I_{x\beta} = 0$, proving the result. □

Exact moment conditions. The optimal temporal schedule does not merely make gains "stationary." It makes the object score orthogonal, in the noise metric, to every admitted drift mode. Expanded by columns h_j of H, the exact moment conditions are, for diagonal $R = \text{diag}(\sigma_n^2)$,

$$\sum_{n=1}^{N} h_j(t_n)\, \frac{s_n}{\sigma_n^2}\, m_n = 0, \qquad j = 1, \ldots, p. \tag{O4.4}$$

Meaning: chopping, interleaving, randomization. This is the theorem behind chopping, interleaving, and randomization:

- *chopping/paired designs* are exact when their signed effective rows annihilate the low-order temporal moments in (O4.4);
- *interleaving* reduces the correlation between object-carrier directions and low-frequency gain modes;
- *random permutation* makes the cross block mean-zero under centered exchangeable rows and gives concentration around zero.

The first two are exact only when the pattern/carrier moment equations actually hold. "Alternating is always optimal" is false.

7 Randomization corollary and flip-boundary rederivation

7.1 Randomization proposition and its obstacle

Proposition 1 (Randomization concentration)**.** *Under bounded centered carrier vectors and a uniformly random permutation, standard matrix concentration bounds*

$$\|I_{x\beta}\|_{\mathrm{op}} = O_{\mathbb{P}}\left(B\sqrt{\frac{\log((K+p)/\delta)}{N}}\right),$$

which in turn bounds the Schur-complement loss by

$$\|I_{xx} - I_{x|\mathrm{gain}}\|_{\mathrm{op}} \le \frac{\|I_{x\beta}\|_{\mathrm{op}}^2}{\lambda_{\min}(I_{\beta\beta})}.$$

Remark 5 (Named obstacle — open lemma). Proposition 1 is a theorem only under explicitly bounded, centered, permutation-sampling assumptions. It is *not* yet a theorem for arbitrary non-negative GI patterns, because $s_n = m_n^\top x$ is scene-dependent and can destroy centering. The named obstacle is the product $s_n m_n$: temporal randomization randomizes a scene-dependent carrier, not a fixed zero-mean row sequence. *Closing this lemma requires the stationary-carrier hypothesis of the companion Part I* [9]; that hypothesis is exactly the condition needed to complete the proof. This remark must remain until the lemma is closed.

7.2 Relation to the existing flip boundary

The paired schedule can spend measurements/references to reduce $I_{x\beta}$; a randomized schedule preserves more distinct object rows but only suppresses the cross block statistically. The correct flip criterion is therefore equality of a task criterion Φ applied to the two Schur complements,

$$\Phi\{I_{x|\mathrm{gain}}^{\mathrm{pair}}\} = \Phi\{I_{x|\mathrm{gain}}^{\mathrm{rand}}\}, \tag{O4.5}$$

under the same total-time and photon accounting. Part I's existing ρ^* flip may be a closed-form instance of this equality; it should be rederived from (O4.2), not merely re-described as a heuristic schedule boundary. The measure-valued/KKT extension is then classical optimal design [13, 14, 15] applied to atoms that carry both a spatial pattern and a temporal slot. The novel content is the correlated-gain information atom and the nuisance-orthogonality geometry.

8 A model-specific temporal operating surface

This section is a deliberate stub. The R26 ruling rejects a universal two-axis "temporal ridge" and retains only a conditional program: a decorrelation-time ratio alone does not create an interior optimum, because gain averaging over a frame is monotone and produces no interior extremum. An operating surface becomes well posed only after a competing mechanism and a full resource budget are fixed — finite total acquisition time, read noise or switching overhead, per-frame photon budget, pattern rank/identifiability, and the exact within-frame gain functional. Accordingly we state the operating map as a future, conditional target and make no ridge claim here.

Conjecture 1 (O2 — resource-conditioned operating surface; FUTURE / conditional)**.** *Fix a specified OU-gain process, a fixed total acquisition time, and a read-noise-plus-shot-noise GI model. Then a matched temporal operating surface exists on a slice of the resource parameters*

$$r = \frac{T_f}{t_c}, \quad \eta = \lambda t_c, \quad R_{\mathrm{tot}} = \frac{T_{\mathrm{total}}}{t_c}, \quad \omega = \frac{t_{\mathrm{switch}}}{t_c}, \quad N/K,$$

together with a read-noise-to-shot-noise ratio. Any ridge, its constant, and even its existence are model-dependent and must be derived only after these preconditions are fixed. The named obstacle is the coupled growth/decline of pattern rank, gain posterior uncertainty, and per-frame photon information: there is no scalar renewal reduction analogous to dead time. This is a conjecture, not a theorem, and it must not enter the title or the abstract until a concrete resource-conditioned law is derived.

9 Reach: gain drift plus dead-time counting

Let the complete latent record contain the gain path, photon arrivals, and detector state. Conditional-score operators compose through nested observation channels by the tower property, so the controlled-score Gramian architecture survives. What does *not* survive is a scalar product law. If a frame's incident rate is $a_n s_n$ and the counter response is nonlinear, the score depends on

$$\frac{\partial}{\partial x} \log p_{\text{count}}(Y_n \mid a_n s_n),$$

so the posterior uncertainty of a_n is weighted by the derivative and curvature of the dead-time count law. High-gain states are preferentially compressed or saturated. Consequently:

- gain uncertainty and dead-time loss are coupled;
- temporal carriers are clipped most strongly exactly when a_n is large;
- a correction trained for an analog linear bucket may become biased under photon-counting saturation;
- the best temporal schedule may shift because extreme-gain frames carry less marginal object information.

The exact O1 identity (O1.2) still holds if the complete score includes both hidden mechanisms. The first composite treatment should derive one explicit Poisson-renewal specialization; it must **not** claim a mechanism-by-mechanism multiplicative efficiency law.

Code and data availability

Numerical checks of the O4.2 Schur formula and the O4.4 orthogonality conditions, and any released code and evidence bundles, are available at [repo URL wording — user decision].

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